Chapter 6. The Fourier Transform on Rd

Wave equation, radial functions, Radon transform

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6.1 Preliminaries

Motivation. In higher dimensions rotations enter alongside translations and dilations, and integration in polar coordinates splits $\mathbb{R}^d$ into radii and directions. These are the tools for radial functions, spherical means and the Radon transform.

6.1.1 Symmetries

Analysis on $\mathbb{R}^d$ is shaped by **translations** $x \mapsto x+h$, **dilations** $x \mapsto \delta x$ ($\delta > 0$) and **rotations**: linear R with $R(x)\cdot R(y) = x\cdot y$, equivalently $|Rx| = |x|$, or $R^t = R^{-1}$; then $\det R = \pm 1$ (**proper** if $+1$). On $\mathbb{R}$ the rotations are $\pm I$; on $\mathbb{R}^2$ they are $z \mapsto e^{i\varphi}z$ and $z \mapsto e^{i\varphi}\bar{z}$ (identifying $\mathbb{R}^2 = \mathbb{C}$). For $d = 3$, Euler angles parametrize the proper rotations.

6.1.2 Integration on $\mathbb{R}^d$

A continuous f is **rapidly decreasing** if $\sup_x |x|^k |f(x)| < \infty$ for all k; then $\int_{\mathbb{R}^d} f = \lim_{N\to\infty} \int_{Q_N} f$ over the cubes $Q_N = \{|x_i| \le N/2\}$ (or balls). It suffices that $\sup |x|^{d+\epsilon}|f(x)| < \infty$; **moderate decrease** means $\epsilon = 1$. For such f:

$$\int f(x+h)\,dx = \int f(x)\,dx, \qquad \delta^d \int f(\delta x)\,dx = \int f(x)\,dx, \qquad \int f(Rx)\,dx = \int f(x)\,dx.$$

Polar coordinates. Writing $x = r\gamma$, $r \ge 0$, $\gamma \in S^{d-1}$, with the surface measure $d\sigma$ on the sphere,

$$\int_{\mathbb{R}^d} f(x)\,dx = \int_{S^{d-1}} \int_0^\infty f(r\gamma)\, r^{d-1}\, dr\, d\sigma(\gamma).$$

For $d = 2$: $\int f = \int_0^{2\pi}\int_0^\infty f(r\cos\theta, r\sin\theta)\, r\, dr\, d\theta$. For $d = 3$: $x = (r\sin\theta\cos\varphi, r\sin\theta\sin\varphi, r\cos\theta)$, $d\sigma = \sin\theta\, d\theta\, d\varphi$. The area of S^{d-1} is $\omega_d = \frac{2\pi^{d/2}}{\Gamma(d/2)}$ (computed from $\int e^{-\pi|x|^2}dx = 1$).

6.2 Elementary theory of the Fourier transform

Motivation. Everything from Chapter 5 extends to $\mathbb{R}^d$ with $x\xi$ replaced by $x\cdot\xi$. The new feature is the interaction with rotations: the Fourier transform commutes with them, so radial functions have radial transforms.

Multi-index notation: $x^\alpha = x_1^{\alpha_1}\cdots x_d^{\alpha_d}$, $(\frac{\partial}{\partial x})^\alpha$, $|\alpha| = \alpha_1 + \cdots + \alpha_d$. $\mathcal{S}(\mathbb{R}^d)$ consists of smooth f with $\sup_x |x^\alpha (\frac{\partial}{\partial x})^\beta f(x)| < \infty$ for all α, β. The Fourier transform is $\hat{f}(\xi) = \int_{\mathbb{R}^d} f(x) e^{-2\pi i x\cdot\xi} dx$.

Proposition 2.1

For $f \in \mathcal{S}(\mathbb{R}^d)$:
(i) $f(x+h) \to \hat{f}(\xi)e^{2\pi i\xi\cdot h}$;
(ii) $f(x)e^{-2\pi i x\cdot h} \to \hat{f}(\xi+h)$;
(iii) $f(\delta x) \to \delta^{-d}\hat{f}(\delta^{-1}\xi)$, $\delta > 0$;
(iv) $(\frac{\partial}{\partial x})^\alpha f(x) \to (2\pi i\xi)^\alpha \hat{f}(\xi)$;
(v) $(-2\pi i x)^\alpha f(x) \to (\frac{\partial}{\partial \xi})^\alpha \hat{f}(\xi)$;
(vi) $f(Rx) \to \hat{f}(R\xi)$ for every rotation R.

Proof. (i)–(v) as in Proposition 1.2 of Chapter 5 (one variable at a time).

(vi) Substitute $y = Rx$: $\int f(Rx)e^{-2\pi ix\cdot\xi}dx = \int f(y)e^{-2\pi iR^{-1}y\cdot\xi}dy = \int f(y)e^{-2\pi iy\cdot R\xi}dy = \hat{f}(R\xi)$, since $R^{-1}y\cdot\xi = y\cdot R\xi$ ($R^{-1} = R^t$). ■

Corollary 2.2

The Fourier transform maps $\mathcal{S}(\mathbb{R}^d)$ to itself.

Proof. By (iv), (v), $\xi^\alpha(\partial_\xi)^\beta\hat{f}$ is the transform of a Schwartz function, hence bounded. ■

Corollary 2.3

The Fourier transform of a radial function ($f(Rx) = f(x)$ for all rotations R) is radial.

Proof. By (vi), $\hat{f}(R\xi) = \hat{f}(\xi)$ for all R. ■

The Gaussian $e^{-\pi|x|^2} = \prod_j e^{-\pi x_j^2}$ is radial and its own Fourier transform (product of the one-dimensional transforms); hence $K_\delta(x) = \delta^{-d/2}e^{-\pi|x|^2/\delta}$ has $\widehat{K_\delta}(\xi) = e^{-\pi\delta|\xi|^2}$ and is a good kernel as $\delta\to 0$.

Theorem 2.4 (Fourier inversion and Plancherel)

If $f\in\mathcal{S}(\mathbb{R}^d)$, then $f(x) = \int_{\mathbb{R}^d}\hat{f}(\xi)e^{2\pi ix\cdot\xi}d\xi$ and $\int|\hat{f}(\xi)|^2d\xi = \int|f(x)|^2dx$. In particular $\mathcal{F}$ is a bijection of $\mathcal{S}(\mathbb{R}^d)$.

Proof. Exactly as in Chapter 5: the multiplication formula $\int f\hat{g} = \int\hat{f}g$ (Fubini), applied with $g = e^{-\pi\delta|x|^2}$, gives inversion at 0 via the good kernels K_δ; translate for general x. For Plancherel, use $h = f * f^\flat$, $f^\flat(x) = \overline{f(-x)}$, with $\widehat{f*g} = \hat{f}\hat{g}$. ■

6.3 The wave equation in $\mathbb{R}^d\times\mathbb{R}$

Motivation. The Fourier transform solves the wave equation explicitly and shows that energy is conserved. In three dimensions the solution is an average over spheres, which yields Huygens' principle: signals propagate sharply. Two dimensions are handled by viewing them as three-dimensional data independent of x_3 ("descent").

Cauchy problem. Find $u(x,t)$ with $\triangle u = \frac{\partial^2u}{\partial t^2}$ (speed $c=1$), $u(x,0) = f(x)$, $\partial_t u(x,0) = g(x)$, $f,g\in\mathcal{S}(\mathbb{R}^d)$.

6.3.1 Solution in terms of Fourier transforms

Transforming in x: $\partial_t^2\hat{u}(\xi,t) = -4\pi^2|\xi|^2\hat{u}(\xi,t)$, so $\hat{u} = \hat{f}(\xi)\cos(2\pi|\xi|t) + \hat{g}(\xi)\frac{\sin(2\pi|\xi|t)}{2\pi|\xi|}$.

Theorem 3.1

A solution of the Cauchy problem for the wave equation is

$$u(x,t) = \int_{\mathbb{R}^d}\Big[\hat{f}(\xi)\cos(2\pi|\xi|t) + \hat{g}(\xi)\frac{\sin(2\pi|\xi|t)}{2\pi|\xi|}\Big]e^{2\pi ix\cdot\xi}d\xi. \tag{6.1}$$

Proof. $\cos(2\pi|\xi|t)$ and $\frac{\sin(2\pi|\xi|t)}{2\pi|\xi|}$ are smooth in ξ (even power series in $|\xi|$) and, with all their t-derivatives, bounded by polynomials in $|\xi|$ for bounded t. Since $\hat f, \hat g$ are rapidly decreasing, we may differentiate under the integral twice in x and t: each term $[\dots]e^{2\pi i x\cdot\xi}$ satisfies $\triangle_x = -4\pi^2|\xi|^2 = \partial_t^2$. At $t=0$, Fourier inversion gives $u = f$, $\partial_t u = g$. ∎

The **energy** of u is $E(t) = \int_{\mathbb{R}^d} \left(\left|\frac{\partial u}{\partial t}\right|^2 + \sum_j \left|\frac{\partial u}{\partial x_j}\right|^2 \right) dx$.

Theorem 3.2

If u is given by (6.1), then $E(t) = E(0)$ for all t.

Lemma 3.3

For $a, b \in \mathbb{C}$ and $\alpha \in \mathbb{R}$: $|a\cos\alpha + b\sin\alpha|^2 + |-a\sin\alpha + b\cos\alpha|^2 = |a|^2 + |b|^2$.

Proof. The vectors $e_1 = (\cos\alpha, \sin\alpha)$, $e_2 = (-\sin\alpha, \cos\alpha)$ form an orthonormal basis of $\mathbb{R}^2 \subseteq \mathbb{C}^2$, and $(a,b)\cdot e_1$, $(a,b)\cdot e_2$ are the coordinates of (a,b) in it; the identity is Pythagoras (expand: cross terms cancel). ∎

Proof of Theorem 3.2. By Plancherel and Proposition 2.1(iv), $\int |\partial_t u|^2 = \int \left| -2\pi|\xi|\hat f \sin(2\pi|\xi|t) + \hat g\cos(2\pi|\xi|t)\right|^2 d\xi$ and $\sum_j \int |\partial_{x_j} u|^2 = \int \left|2\pi|\xi|\hat f\cos(2\pi|\xi|t) + \hat g\sin(2\pi|\xi|t)\right|^2 d\xi$ (since $\sum_j |2\pi i\xi_j|^2 = 4\pi^2|\xi|^2$). By Lemma 3.3 with $a = 2\pi|\xi|\hat f$, $b = \hat g$, $\alpha = 2\pi|\xi|t$, the sum of the integrands is $4\pi^2|\xi|^2|\hat f|^2 + |\hat g|^2$, independent of t. ∎

Uniqueness [Supplement]

Conservation of energy (for any solution with Schwartz-type decay, by integrating by parts as in Theorem 2.3 of Chapter 5: $E'(t) = 2\,\mathrm{Re}\int(\partial_t u\,\overline{\partial_t^2 u} + \sum \partial_j u\,\overline{\partial_j \partial_t u}) = 2\,\mathrm{Re}\int \partial_t u\,\overline{(\partial_t^2 u - \triangle u)} = 0$) shows that a solution with zero data has $E \equiv 0$, so u is constant in x and t, hence 0. Thus (6.1) is the only such solution.

6.3.2 The wave equation in $\mathbb{R}^3 \times \mathbb{R}$

The **spherical mean** of f is $M_t(f)(x) = \frac{1}{4\pi}\int_{S^2} f(x - t\gamma)\, d\sigma(\gamma)$, the average of f over the sphere of radius t centred at x.

Lemma 3.4

If $f \in \mathcal{S}(\mathbb{R}^3)$ and t is fixed, then $M_t(f) \in \mathcal{S}(\mathbb{R}^3)$; moreover $M_t(f)$ is C^∞ in t and each t-derivative lies in $\mathcal{S}(\mathbb{R}^3)$.

Proof. For $|x| \geq 2|t|$ and $\gamma \in S^2$, $|x - t\gamma| \geq |x|/2$, so $|f(x-t\gamma)| \leq C_N(1+|x|)^{-N}$ and $M_t(f)$ is rapidly decreasing. Derivatives in x and t pass under the integral and are spherical means of derivatives of f (times polynomials in γ), so they are rapidly decreasing too. ∎

Lemma 3.5

$\frac{1}{4\pi}\int_{S^2} e^{-2\pi i\xi\cdot\gamma} d\sigma(\gamma) = \frac{\sin(2\pi|\xi|)}{2\pi|\xi|}$.

Proof. The left side is radial in ξ (rotation invariance of $d\sigma$), so take $\xi = (0,0,\rho)$, $\rho = |\xi|$. In spherical coordinates

$$\frac{1}{4\pi}\int_0^{2\pi}\int_0^{\pi} e^{-2\pi i\rho\cos\theta}\sin\theta\, d\theta\, d\varphi = \frac{1}{2}\int_{-1}^{1} e^{-2\pi i\rho u}du = \frac{\sin(2\pi\rho)}{2\pi\rho}.$$

■

Consequently (Fubini),

$$\widehat{M_t(f)}(\xi) = \hat{f}(\xi)\frac{\sin(2\pi|\xi|t)}{2\pi|\xi|t}. \tag{6.2}$$

Theorem 3.6 (Kirchhoff)

For $d = 3$, the solution of the Cauchy problem is $u(x,t) = \frac{\partial}{\partial t}(tM_t(f)(x)) + tM_t(g)(x)$.

Proof. By Theorem 3.1 with $f = 0$, the solution with data $(0, g)$ is

$$u_1 = \int \hat{g}(\xi)\frac{\sin(2\pi|\xi|t)}{2\pi|\xi|}e^{2\pi ix\cdot\xi}d\xi = t\int \hat{g}\frac{\sin(2\pi|\xi|t)}{2\pi|\xi|t}e^{2\pi ix\cdot\xi}d\xi = tM_t(g)(x),$$

by (6.2) and inversion. With data $(f, 0)$, $u_2 = \int \hat{f}\cos(2\pi|\xi|t)e^{2\pi ix\cdot\xi}d\xi = \frac{\partial}{\partial t}\left(t\int \hat{f}\frac{\sin(2\pi|\xi|t)}{2\pi|\xi|t}e^{2\pi ix\cdot\xi}d\xi\right) = \partial_t(tM_t(f))$. Superpose: $u = u_1 + u_2$. ■

Huygens' principle. In $d = 1$, $u(x,t)$ (d'Alembert) depends on the data in the whole interval $[x-t, x+t]$, but only on the endpoints if $g = 0$. In $d = 3$, $u(x,t)$ always depends only on the data on the *sphere* $|y - x| = t$: a disturbance localized near a point is felt at distance t exactly at time t and then passes (sharp signals; light and sound propagate cleanly).

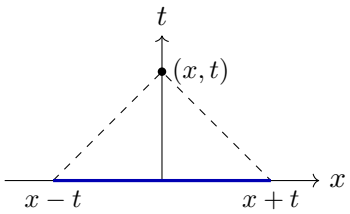


Figure. *Domain of dependence in $d = 1$: $u(x,t)$ depends on the data in $[x-t, x+t]$; in $d = 3$ only on the sphere of radius t.*

6.3.3 The wave equation in $\mathbb{R}^2 \times \mathbb{R}$: descent

For $f \in \mathcal{S}(\mathbb{R}^2)$ let $\tilde{M}_t(f)(x) = \frac{1}{2\pi}\int_{|y|\le 1} f(x - ty)(1 - |y|^2)^{-1/2}dy$.

Theorem 3.7

A solution of the Cauchy problem in two dimensions with data $f, g \in \mathcal{S}(\mathbb{R}^2)$ is

$$u(x,t) = \frac{\partial}{\partial t}(t\tilde{M}_t(f)(x)) + t\tilde{M}_t(g)(x). \tag{6.3}$$

Proof.

Step 1 (descent idea). Let $\tilde{f}(x_1, x_2, x_3) = f(x_1, x_2)$; a solution of the 3-dimensional problem with data independent of x_3 is independent of x_3 and solves the 2-dimensional problem. Projecting the

upper and lower hemispheres $\gamma = (y, \pm\sqrt{1-|y|^2})$, $|y| \le 1$, onto the unit disc, $d\sigma = (1-|y|^2)^{-1/2}dy$ on each, so $M_t(\tilde f)(x,x_3) = \frac{2}{4\pi}\int_{|y|\le 1} f(x-ty)(1-|y|^2)^{-1/2}dy = \tilde M_t(f)(x)$.

Step 2 (truncation). Since $\tilde f \notin \mathcal{S}(\mathbb{R}^3)$, fix $T>0$ and $\eta \in \mathcal{S}(\mathbb{R})$ with $\eta(x_3)=1$ for $|x_3|\le 3T$, and let $\tilde f^\flat = f(x_1,x_2)\eta(x_3)$, $\tilde g^\flat$ likewise. By Theorem 3.6, $u^\flat = \partial_t(tM_t(\tilde f^\flat)) + tM_t(\tilde g^\flat)$ solves the 3-dimensional problem with these data. For $|x_3|\le T$ and $|t|\le T$, the spheres of radius t around (x,x_3) stay in $|x_3|\le 2T$, where $\tilde f^\flat = \tilde f$, so $u^\flat(x,x_3,t)$ equals (6.3). It is independent of x_3 there, so $\partial_{x_3}^2 u^\flat = 0$ and (6.3) solves the 2-dimensional equation for $|t|<T$. T is arbitrary. ∎

In two dimensions $u(x,t)$ depends on the data on the whole disc of radius t, so Huygens' principle fails (ripples on water linger).

6.4 Radial symmetry and Bessel functions

Motivation. For radial $f(x) = f_0(|x|)$ the transform is radial, $\hat f(\xi) = F_0(|\xi|)$. In dimensions 1 and 3 the relation between f_0 and F_0 is elementary; in dimension 2 it involves the Bessel function J_0.

- $d=1$ (radial = even): $F_0(\rho) = \int f(x)e^{-2\pi i x\rho}dx = 2\int_0^\infty \cos(2\pi\rho r) f_0(r)\,dr$.
- $d=3$: by polar coordinates and Lemma 3.5,

$$\begin{aligned} F_0(\rho) &= \int_0^\infty f_0(r)\Big(\int_{S^2} e^{-2\pi i r\gamma\cdot\xi}d\sigma(\gamma)\Big)r^2dr \\ &= \int_0^\infty f_0(r)\frac{2\sin(2\pi\rho r)}{\rho r}r^2dr = 2\rho^{-1}\int_0^\infty \sin(2\pi\rho r) f_0(r)\,r\,dr. \end{aligned}$$

- $d=2$: the **Bessel function** of order $n\in\mathbb{Z}$ is $J_n(\rho) = \frac{1}{2\pi}\int_0^{2\pi} e^{i\rho\sin\theta}e^{-in\theta}d\theta$, the n-th Fourier coefficient of $e^{i\rho\sin\theta}$, so $e^{i\rho\sin\theta} = \sum_n J_n(\rho)e^{in\theta}$. Taking $\xi = (0,-\rho)$,

$$F_0(\rho) = \int_0^{2\pi}\int_0^\infty f_0(r)e^{2\pi i r\rho\sin\theta}r\,dr\,d\theta = 2\pi\int_0^\infty J_0(2\pi r\rho) f_0(r)\,r\,dr.$$

In general F_0 is given by a Bessel function of order $d/2-1$; for odd d these are elementary, for even d not.

6.5 The Radon transform and some of its applications

Motivation. X-ray scans measure integrals of the attenuation coefficient along lines. Reconstructing a function from its integrals over lines (in $\mathbb{R}^2$) or planes (in $\mathbb{R}^3$) is possible because the one-dimensional Fourier transform of the Radon transform is a slice of the full Fourier transform.

6.5.1 The X-ray transform in $\mathbb{R}^2$

A beam of intensity I_0 traversing an object with attenuation coefficient ρ along a line L emerges with intensity $I = I_0 e^{-\int_L \rho}$. So measuring I for all lines gives the **X-ray transform** $X(\rho)(L) = \int_L \rho$. **Reconstruction problem:** recover ρ from $X(\rho)$ (uniqueness and an inversion formula). In $\mathbb{R}^2$ the inversion formula involves a Hilbert transform; we treat the simpler 3-dimensional analogue.

6.5.2 The Radon transform in $\mathbb{R}^3$

Planes in $\mathbb{R}^3$ are $\mathcal{P}_{t,\gamma} = \{x : x \cdot \gamma = t\}$, $t \in \mathbb{R}$, $\gamma \in S^2$ (and $\mathcal{P}_{t,\gamma} = \mathcal{P}_{-t,-\gamma}$). For $f \in \mathcal{S}(\mathbb{R}^3)$ choose e_1, e_2 with $\{e_1, e_2, \gamma\}$ orthonormal and set

$$\mathcal{R}(f)(t,\gamma) = \int_{\mathcal{P}_{t,\gamma}} f = \int_{\mathbb{R}^2} f(t\gamma + u_1 e_1 + u_2 e_2)\, du_1\, du_2.$$

Proposition 5.1

For $f \in \mathcal{S}(\mathbb{R}^3)$ and each γ, $\mathcal{R}(f)(t,\gamma)$ does not depend on the choice of e_1, e_2, and $\int_{-\infty}^{\infty} \mathcal{R}(f)(t,\gamma)\, dt = \int_{\mathbb{R}^3} f(x)\, dx$.

Proof. Another choice e_1', e_2' differs by a rotation (or reflection) of $\mathbb{R}^2$, under which $\int_{\mathbb{R}^2}$ is invariant. For the identity, the linear change of variables $x = t\gamma + u_1 e_1 + u_2 e_2$ is a rotation of $\mathbb{R}^3$, so $\int_{\mathbb{R}^3} f = \int_{\mathbb{R}} \int_{\mathbb{R}^2} f(t\gamma + u_1 e_1 + u_2 e_2)\, du\, dt$ (Fubini for rapidly decreasing functions). ∎

Lemma 5.2

For $f \in \mathcal{S}(\mathbb{R}^3)$ and fixed γ, $\mathcal{R}(f)(\cdot,\gamma) \in \mathcal{S}(\mathbb{R})$, and $\widehat{\mathcal{R}(f)}(s,\gamma) = \hat{f}(s\gamma)$ (one-dimensional transform in t on the left, three-dimensional on the right).

Proof. $(1+|t|)^N |f(t\gamma + u)| \le C(1+|u|)^{-3}$ since $|t\gamma + u|^2 = t^2 + |u|^2$, so $\mathcal{R}(f)$ decays rapidly in t; derivatives in t are Radon transforms of directional derivatives of f. By Proposition 5.1 applied to $f(x)e^{-2\pi i s x\cdot\gamma}$ (which equals $e^{-2\pi i s t} f(x)$ on $\mathcal{P}_{t,\gamma}$), $\int \mathcal{R}(f)(t,\gamma) e^{-2\pi i s t} dt = \int_{\mathbb{R}^3} f(x) e^{-2\pi i s x\cdot\gamma} dx = \hat{f}(s\gamma)$. ∎

Corollary 5.3

If $f, g \in \mathcal{S}(\mathbb{R}^3)$ and $\mathcal{R}(f) = \mathcal{R}(g)$, then $f = g$.

Proof. By Lemma 5.2, $\hat{f}(s\gamma) = \hat{g}(s\gamma)$ for all s, γ, i.e. $\hat{f} = \hat{g}$; apply Fourier inversion. ∎

The **dual Radon transform** of $F(t,\gamma)$ is $\mathcal{R}^*(F)(x) = \int_{S^2} F(x\cdot\gamma, \gamma)\, d\sigma(\gamma)$: it averages F over all planes through x.

Theorem 5.4 (Radon inversion)

If $f \in \mathcal{S}(\mathbb{R}^3)$, then $\triangle(\mathcal{R}^*\mathcal{R}(f)) = -8\pi^2 f$.

Proof. By Lemma 5.2 and one-dimensional inversion, $\mathcal{R}(f)(t,\gamma) = \int_{\mathbb{R}} \hat{f}(s\gamma) e^{2\pi i t s} ds$, so $\mathcal{R}^*\mathcal{R}(f)(x) = \int_{S^2} \int_{\mathbb{R}} \hat{f}(s\gamma) e^{2\pi i x\cdot\gamma s} ds\, d\sigma(\gamma)$. Applying $\triangle$ (under the integral) multiplies the integrand by $-4\pi^2 s^2 |\gamma|^2 = -4\pi^2 s^2$. The change $(s,\gamma) \mapsto (-s,-\gamma)$ shows that the parts $s<0$ and $s>0$ are equal, so

$$\triangle(\mathcal{R}^*\mathcal{R}(f))(x) = -8\pi^2 \int_{S^2} \int_0^{\infty} \hat{f}(s\gamma) e^{2\pi i x\cdot s\gamma} s^2 ds\, d\sigma(\gamma) = -8\pi^2 f(x),$$

the last step being Fourier inversion written in polar coordinates. ∎

So $f = -\frac{1}{8\pi^2} \triangle \mathcal{R}^*\mathcal{R}(f)$: integrals over planes determine f explicitly, the mathematical basis of tomography.